LECTURE 2

1. Units

$E = \frac{3}{2} k_B t_K$ $\boxed{T = k_B t_K}$

↑

human unit

$k_B = 1.4 \times 10^{-23}$ J/K

2. Entropy

S_{CARNOT}

Boltzman Entropy:

$\downarrow$

$S = -\sum_i p_i \log p_i = \frac{1}{k_B} S_{CARNOT}$

Engine Entropy

$\downarrow$

$\downarrow$

Statistical Distribution

- Boltzman:

$S = k \ln W$

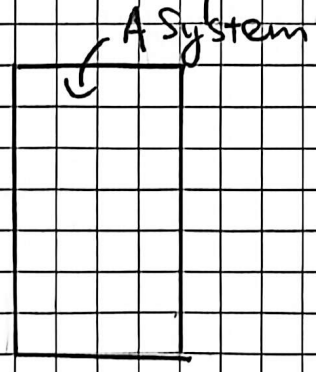
↑

number of microscopic arrangement

- Engine:

$dS = -\frac{dQ}{T}$ → reversible heat transfer

3. Thermal Equilibrium



State: i ⟶ Probability: $P(i)$

$\downarrow$

Energy: E_i

$\sum_i P(i) = 1$

Average Energy of the system $= \sum_i P(i) E_i = \langle E \rangle$ parameter

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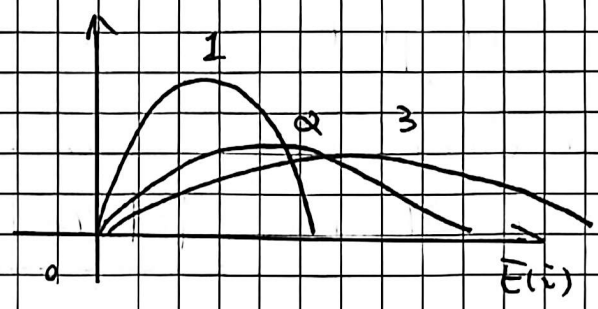
$p(i, \langle E \rangle)$

$\downarrow$

$\sum_i p(i, \langle E \rangle) E_i = \langle E \rangle$



$1 \to 2 \to 3$

$E \uparrow$

$$S(E) = -\sum_i p(i, E) \log p(i, E)$$

4. One Parameter Family

$$\begin{cases} \Delta E = \frac{\partial E}{\partial S} \Delta S \\ \Delta S = \frac{\partial S}{\partial E} \Delta E \end{cases}$$

$\frac{\partial E}{\partial S} \to T$

$$\boxed{dE = T\,dS \iff dS = \frac{dE}{T}}$$

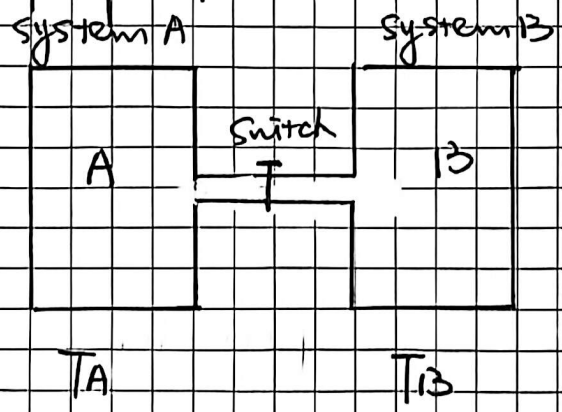
$$T = \frac{\partial E}{\partial S} > 0$$

$$\boxed{dE = t_K\, dS_{CARNOT} \iff dS_{CARNOT} = \frac{dE}{t_K}}$$

since $\begin{cases} S = \frac{1}{K_B} S_{CARNOT} \\ T = K_B t_K \end{cases}$

$T > 0 \Rightarrow \frac{\partial E}{\partial S} > 0 \Rightarrow$ more spread for the diagram

5. Temperature



Energy will flow from system with higher temperature to lower temperature

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$$S = S_A + S_B$$

since $\begin{cases} \text{probability} \rightarrow \text{multiplied} \\ S \text{ is logrithm} \rightarrow \text{addition} \end{cases}$

☆ 1st law of thermaldynamics (Energy Conservation)

$$dE_A = -dE_B$$

☆ 2nd law of thermal dynamics

$$dS_A + dS_B > 0 \iff dS > 0$$

$$dE_A = T_A dS_A$$

$$dE_B = T_B dS_B$$

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